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Analysis of hybrid Nd:Yag laser-MAG arc welding processes

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Abstract

In the hybrid laser-arc welding process, a laser beam and an electric arc are coupled in order to combine the advantages of both processes: high welding speed, low thermal load and high depth penetration thanks to the laser; less demanding on joint preparation/fit-up, typical of arc welding. Thus the hybrid laser-MIG/MAG (Metal Inert or Active Gas) arc welding has very interesting properties: the improvement of productivity results in higher welding speeds, thicker welded materials, joint fit-up allowance, better stability of molten pool and improvement of joint metallurgical quality. The understanding of the main relevant involved physical processes are therefore necessary if one wants for example elaborate adequate simulations of this process. Also, for an efficient use of this process, it is necessary to precisely understand the complex physical phenomena that govern this welding technique. This paper investigates the analysis of the effect of the

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main operating parameters for the laser alone, MAG alone and hybrid Laser/MAG welding processes. The use of a high speed video camera allows us to precisely characterize the melt pool 3D geometry such as the measurements of its depression and its length and the phenomena occurring inside the melt pool through keyhole-melt pool-droplet interaction. These experimental results will form a database that is used for the validation of a three-dimensional thermal model of the hybrid welding process for a rather wide range of operating parameters where the 3-D geometry of the melt pool is taken into account.

Keywords

Hybrid laser/arc; Welding; 3D modelling

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Nomenclature

- C
- Lagrange multiplier
- f
- frequency (Hz)
- g
- gravity (ms^{-2})
- I
- current (A)
- m_d

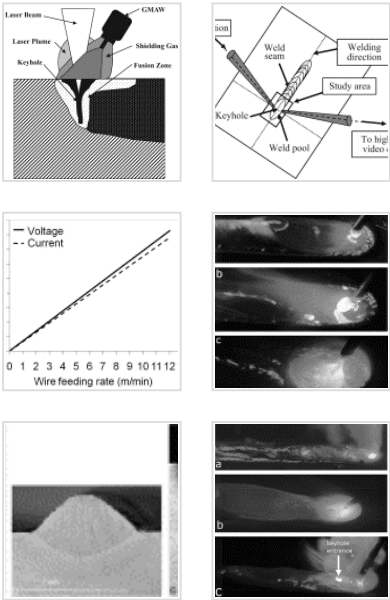
7. Future work

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mean mass of a droplet (kg)

P_{arc}

arc power (W)

P_{surf}

arc pressure (Pa)

P_d

droplet pressure (Pa)

P_{laser}

laser power (W)

R_d

mean droplet radius (m)

r_w

radius of the filler wire (m)

r_0

radius of surface heat flux (m)

r_1

radius of arc pressure (m)

S_{reinf}

weld reinforcement surface (m²)

T

temperature (K)

U

voltage (V)

U_0
welding speed (ms^{-1})

V_d
droplet speed (ms^{-1})

V_w
filler wire speed (ms^{-1})

Greek symbols

ρ
mass density (kgm^{-3})

σ
coefficient of surface tension (Nm^{-1})

η
arc efficiency

ν
droplet frequency (Hz)

φ_{surf}
surface heat flux (Wm^{-2})

Φ
reinforcement height (m)

1. Introduction

Hybrid welding, which combines the techniques of laser welding and arc welding, was originally invented for combined laser-Tungsten Inert Gas (TIG) welding [1]. During hybrid welding, the laser beam creates a vapour capillary, the keyhole, where the beam

energy is deposited throughout the work piece depth, which enables a deep penetration welding effect. The feeding wire is melted by the electric arc created between the MIG/MAG-wire and the work piece. Both the arc and the keyhole create a plasma, thus producing a highly complex physical situation. As two techniques are combined, the number of process parameters is rather large and therefore difficult to optimize in order to achieve the desired weld. A general review and a complete bibliography of hybrid processes can be obtained in the paper of Bagger and Olsen [2].

Once the mutual interaction between the two energy sources is optimized, the combination of the two processes opens large areas of applications and increases its capabilities: the arc welding process, characterized by relatively lower power density and wider process zone, gives a wide bead, thus enhancing the joint's root bridging ability and enlarging the manufacturing tolerances for joint preparation. The laser beam process, characterized by higher localized power density, leads to a deeper penetration. Thus in hybrid Gas Metal Arc (GMA)-laser beam welding, a wide and deep bead is achieved with higher welding speeds compared with the (GMAW) process by its own. As a result, the Gas Metal Arc Welding industrialization of that process has now become a real objective, which can be facilitated for example, by the development of a more efficient hybrid head [3], and/or by a rather complete mastering of the process through basic understanding of it.

Deep penetration allows for the welding of rather thick materials even at high welding speeds enabling the filling of unavoidable gaps of the joints, with possible improved metallurgical properties and controlled geometry of the weld seam [2], [4]. It is known that the strength of a weld seam is defined by many parameters. Its geometry is one of these crucial parameters that have to be controlled or even optimized if one wants to minimize the effect of the induced stress field generated during the loading cycles of these welds. Similarly the metallurgical properties of these welds also depend on the thermal cycles induced by the involved heat sources, the arc and the laser, which are used during this process [5]. So far, most of the hybrid laser-MIG/MAG welding research is focusing on how to effectively combine a laser welding process with an arc welding

process together by experimental methods. These studies have an objective to analyse the influence of various operating parameters on the weld joint shapes. It has been shown that the bead width is mainly controlled by the arc power and consequently the mode of transfer, whereas laser power influences the penetration [5], [6], [7]. The distance between the arc and the laser, as well as the laser beam focal position are also key factors that affect the penetration depth and bead shape [6], [8], [9], [10]. In general, a distance between the two sources smaller than 5 mm is recommended. Kah et al. [11] discussed the choice of the arc-torch direction and the distance between laser and arc. This decision should be made case by case and depends on various parameters, such as the laser beam properties available, the material, arc source parameters and the function of filler wire (to improve metallurgy, to fill up air gap or form weld bead). The weld penetration is deeper when smaller inclinations for MIG torch are used. Casalino has observed deeper penetration using a 45° angle rather than 60° in MIG-laser CO₂ hybrid welding of an Al-Mg alloy. The stability and efficiency of the laser-arc process are influenced by the shielding gas. Tani et al. [12] recommend a 30 up to 40% helium content in hybrid CO₂ laser-GMAW welding of AISI 304 stainless steel. A pulsed/spray transfer mode should be preferred in order to achieve a stable and repeatable process [6]. Other authors have also studied the hybrid laser-arc welding of dissimilar metals [13] and the hybrid welding of aluminum [14].

It should be mentioned that most of these works are essentially based on macrograph analysis. However, observations of the melt pool hydrodynamics using a high speed video camera in the visible range for the melt pool surface analysis [15], [16] or by X-ray radiography [17] for the internal melt pool behaviour give very interesting insights on the physical phenomena and help in understanding the complex interaction between the many parameters involved in this process.

A numerical approach of the hybrid welding was given by Zhou and Tsai [18] and Wouters [19]. The GMA welding process alone has also been previously studied by using a 3D heat and fluid flow model of a melt pool [20]. We will use a similar approach for our hybrid welding description. Cho and Na [21] present a whole modelling of hybrid welding where

he uses the VOF method to track the movement of free surfaces. The Navier–Stokes equation, heat and species conservation are solved for melt pool description. For the behaviour of the capillary, surface tension and recoil pressure are taken into account. The effects of multiple reflections on the walls of the capillary function of time are also taken into account in calculating the melt flow. However this work was only numerical, hence no comparison with the resulting experimental is presented.

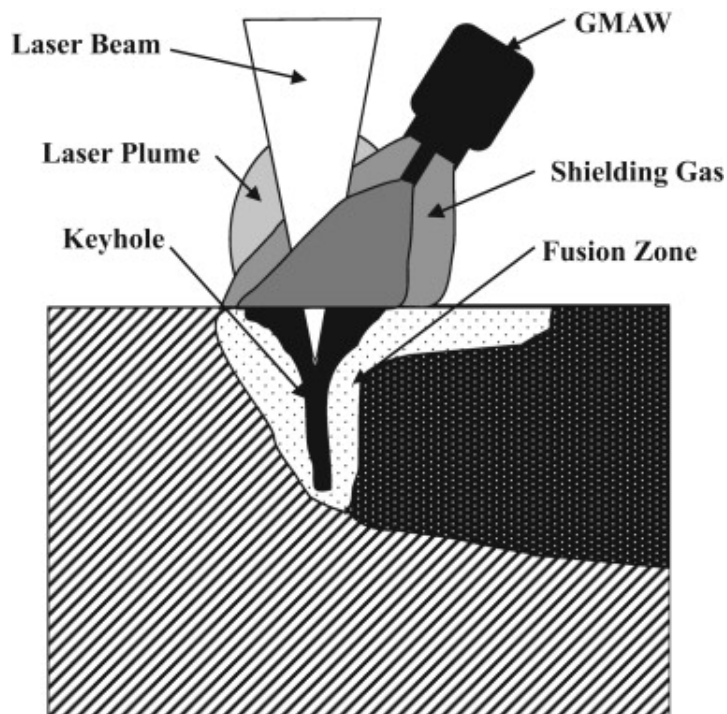
In the first part of this paper, we will present the different experimental results we obtained on the study of hybrid welding, mainly by an extensive use of a high speed video camera that allowed for a better understanding of the phenomena induced during laser, MAG and hybrid laser-MAG processes. These new experimental results complete some previous experimental results and give a complete set of data that must be taken into account and reproduced in the corresponding simulations. The numerical approach will be described in the second part of this paper with corresponding results. We use a numerical modelling of hybrid laser-GMA welding in order to reproduce the 3D geometry of the weld reinforcement observed at the top and the melt pool surface depression observed under the arc. Our final goal here is to define a numerical tool that is rather easy to run and precise enough to reproduce these geometrical observations that appear to be decisive for a correct determination of the induced thermal field, once the main operating parameters are defined. These simulations should allow us to determine in a second step the thermal cycles induced inside the material for estimating the resulting metallurgical transformations. Therefore, these simulations that have been validated with a set of experimental observations will be discussed in the last part of this paper and will allow us to define the corresponding parameters related to arc and laser energy deposition and resulting applied pressures at the melt pool surface.

2. Experimental set-up

Experiments were carried out with a CW Nd:Yag 4006D TRUMPF Laser, with a maximum power of 4kW. The laser beam was delivered through a 600µm optical fibre. The welding head images the exit of the optical fibre by using a collimating and focusing lens that

generates a focal spot of 600 μ m diameter. The focal spot intensity distribution, which was analysed by using a PROMETEC beam analyser, can be considered as uniform.

The arc torch is a DIGI WAVE 400 model from SAFT PRO (Air Liquide Welding France). It delivers up to 400A current for different configurations of control of current and tension of the arc. Welding current and arc voltage traces were recorded via a transient recorder. Arcal21 shielding gas flowing at 20L/min was used (92% Ar and 8% of CO₂). Of course, it is possible to vary all the geometric parameters of the torch such as the position and the wire angle with the vertical laser beam (see Fig. 1).

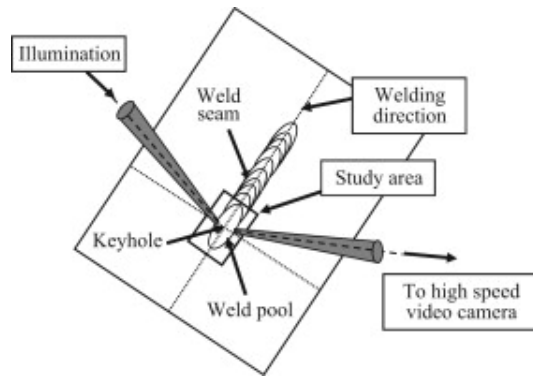


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Fig. 1. Scheme of hybrid laser – MIG/MAG process.

The dynamics of the melt pool was analysed with a CMOS high speed camera (PHOTRON IMAGER FASTCAM APX RS 3000) with a maximum frame rate of 100kHz. However, 10kHz

was sufficient for these configurations. Lateral observations of the melt pool geometry were realized by placing this camera on the side, perpendicularly to the welding displacement, with the optical axis of the camera at an angle of 45° with the vertical of the incident laser beam (see Fig. 2).



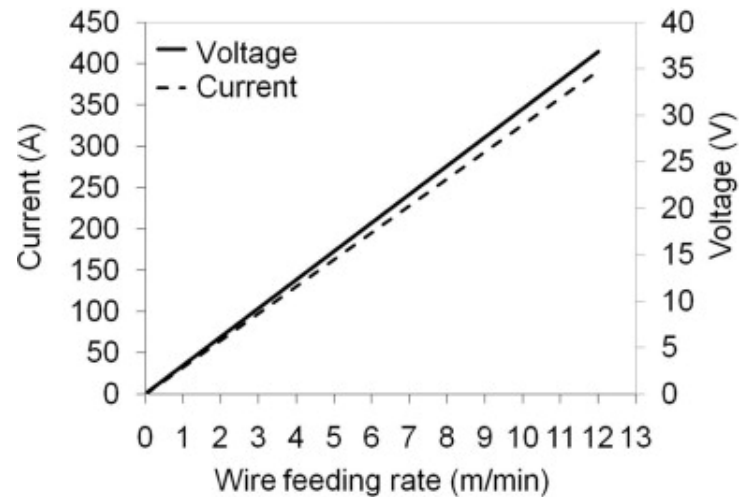
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Fig. 2. Schematic representation of the high speed camera disposition.

The work piece studied here consists in a steel S355 plate of 20mm thick (S355 composition 1.1% Mn, 0.003% Si, 0.014% P, 0.004% S, 0.03% Cu, 0.076% C and 0.005% N). Fusion lines were carried with a G4 Si1 filler wire of 1.2mm diameter (1.5% Mn, 1% Si 0.1% C). After welding, cross sections of the weld beads were ground, polished and etched according to the standard procedures. All these sections were cut midway from the welded specimen. The bead dimensions corresponding to each welding parameter were measured using a stereo microscope. Partial penetration welds were made using the following operating parameters:

- Maximum incident laser power was 4kW.
- Welding speed was progressively increased from 0.5m/min to few m/min until the visual aspect of weld seam was irregular.

- Wire speed (from 2 to 12m/min) and current and voltage were imposed by the synergy of the welding machine (Fig. 3). One must recall that this synergy (U - I relation as a function of wire speed) does not depend on the welding speed.



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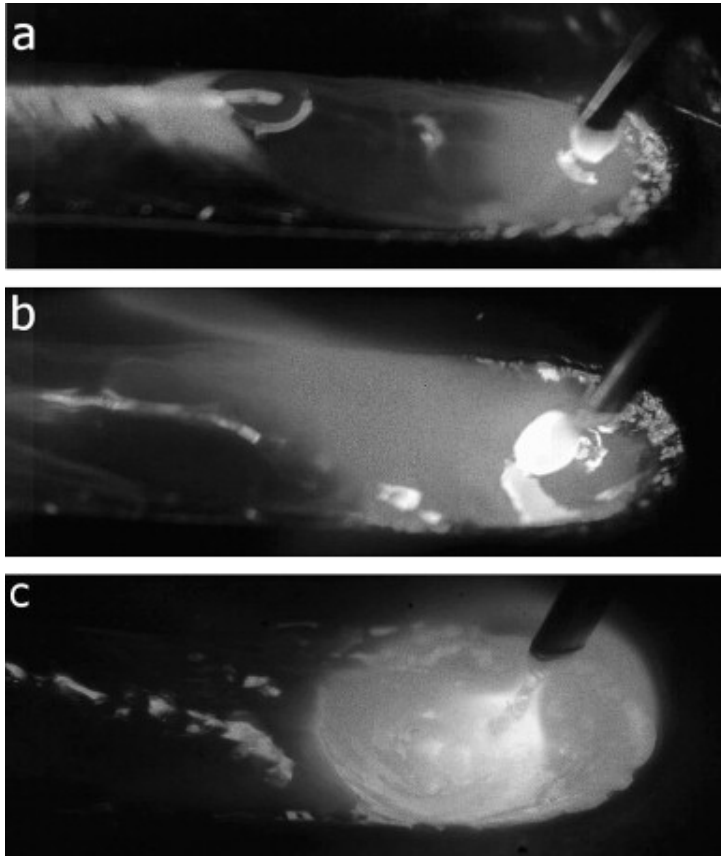
Fig. 3. Voltage and current synergy obtained with the parameters of our study (S355 steel, 1.2mm diameter wire, shielding gas: 92% Ar – 8% CO₂).

- Arc torch inclination of 30° from the vertical laser beam was fixed.
- Laser beam was positioned behind the MAG torch.
- Distance between filler wire (MAG) and laser beam was set to 2mm.

3. Experimental observations

3.1. Metal transfer modes

It is well-known that the wire transfer mode is determined by the heat input and the used shielding gas. In our case, the short-circuiting transfer occurs between 100 and 200A, the globular mode between 200 and 250A and the spray mode from 250 to 400A. The observations using a fast video camera confirm the 3 metal transfer modes according to the current used (Fig. 4).

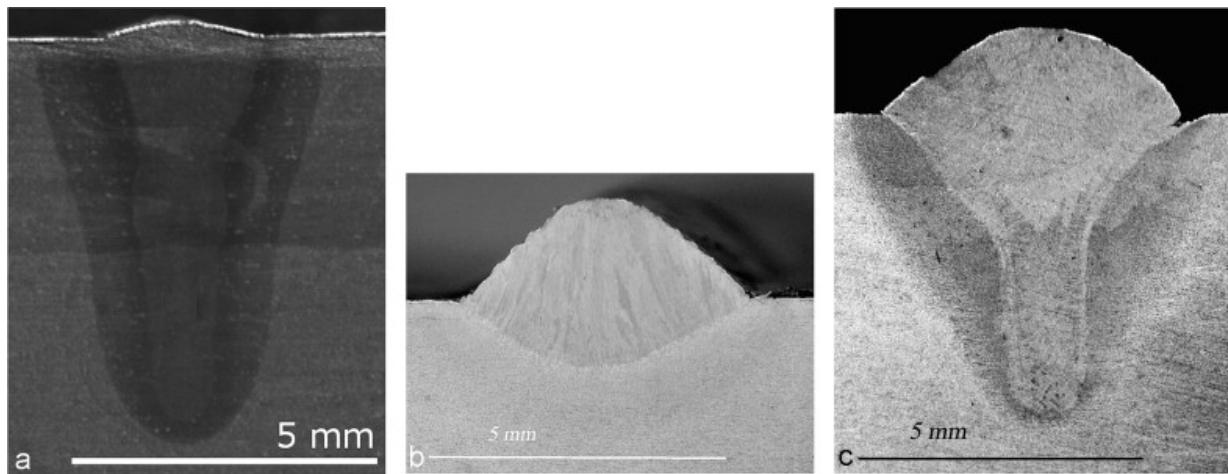


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Fig. 4. Metal transfer in MAG process: (a) short-circuiting ($V_w=3$ m/min), (b) globular ($V_w=6$ m/min), (c) spray ($V_w=11$ m/min).

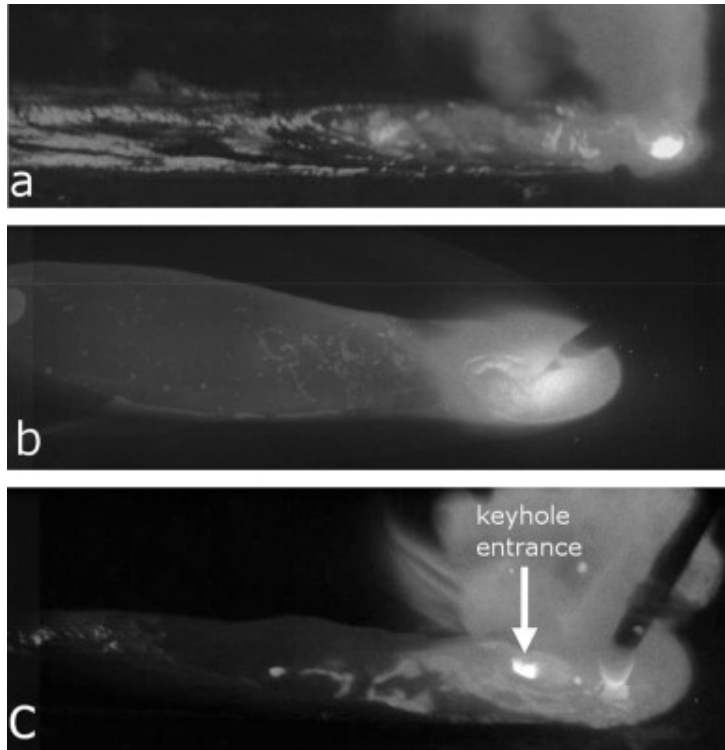
3.2. Geometry

Examples of typical cross sections of weld seams for laser, MAG and Hybrid conditions are shown in Fig. 5. Moreover, Fig. 6 shows the corresponding views of the melt pool observed using the high speed video camera. Due to the rather low welding speeds used here (typically less than 5 m/min), the cross section of a weld with the laser alone shows the well-known nail head shape and the surface melt pool shows the characteristic humps observed around the keyhole rim for these welding conditions.



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Fig. 5. Weld bead obtained for: (a) laser alone: welding speed 1 m/min, laser power 4kW, (b) MAG alone: welding speed 1 m/min, wire speed 5m/min, current 200A, voltage 17.6V and (c) hybrid Laser-MAG: welding speed 1 m/min, laser power 4kW, wire speed 5m/min, current 200A, voltage 17.6V.



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Fig. 6. Examples of images extracted from high speed videos: (a) laser alone: welding speed 1 m/min, laser power 4kW, (b) MAG alone: welding speed 1 m/min, wire speed 5 m/min, current 200A, voltage 17.6V and (c) hybrid laser-MAG: welding speed 1 m/min, laser power 4kW, wire speed 5 m/min, current 200A, voltage 17.6V.

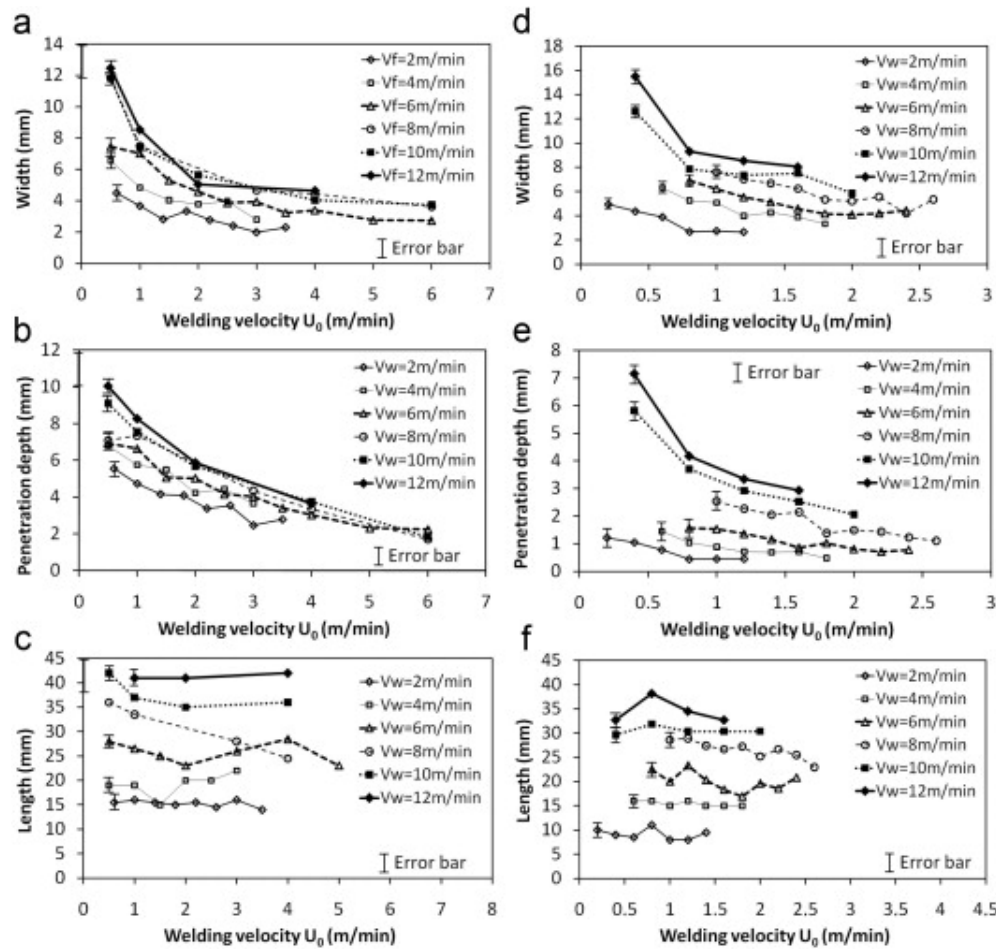
These effects result from the induced melt flow that is dragged upwards along the keyhole wall by the friction forces due to vapour expansion out of the keyhole [22]. The MAG cross section shows the characteristic height seam reinforcement, with a penetration depth that is much lower than in a laser case.

The weld bead shows a rather complex structure where the depressed zone, resulting from the impingement of the droplets flow and the arc pressure area (bright zone), is

followed by the weld reinforcement that is melted along a rather large distance. Finally for the hybrid process, one can observe a greater keyhole penetration compared to the laser alone and the geometry of the reinforcement similar to the one obtained with the MAG process alone.

In that case, the keyhole aperture is located inside the melt pool surface just behind the arc centreline in agreement with the arc trailing position.

Measurements carried out on macrographs (Fig. 7) show that the penetration depth is greater in the hybrid process than in the MAG process. More precisely, the penetration depth ranges from 2 to 10mm in hybrid and from 1 to 7mm in MAG. The weld pool length is also longer in the hybrid process than in the MAG process. It varies from 14 to 43mm in the hybrid process compared to 10–38mm in MAG. Besides, the melt pool width is larger in the MAG process than in the hybrid process. In MAG, the width ranges from 2.5 to 15.5mm, whereas in hybrid, it varies from 2.5 to 12.5mm. As expected, the width and penetration depth decrease when the welding speed increases, and the three characteristic melt pool dimensions (width, length and deep penetration) increase when the wire feeding rate increases. The weld pool length does not seem to vary with the welding speed.



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Fig. 7. Width, penetration and length of the melt pool obtained with hybrid Laser-MAG (a–b–c) and MAG alone (d–e–f) as a function of welding speed, for different wire speeds.

Another point concerns the bead surface reinforcement S_{reinf} defined by the feeding wire, which should depend on the filler wire speed V_w , its wire radius r_w and the welding speed U_0 with the following relation:

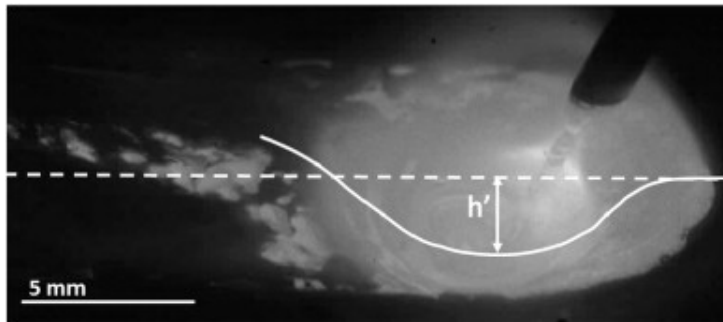
$$U_0 S_{reinf} = V_w \pi r_w^2 \quad (1)$$

In fact, one can observe that the average difference between the calculated (Eq. (1)) and the measured value of the reinforcement is about 13% and 19% for MAG alone and hybrid laser-MAG, respectively. The experimental bead surface reinforcements are always larger than the calculated one and this difference is larger for the hybrid process than for the MAG one. This difference can be explained by the presence of reinforcement that is always observed even during laser welding alone, although there is no material addition in that case. The origin of this reinforcement is likely the effect of the obvious macroscopic deformation of the welded part resulting from the depth gradient of energy deposition inside the material. This effect is particularly important in case of the hybrid process, because of the difference of level and location of the energy input between the two processes (at the surface for MAG compared to surface and volume for the hybrid process).

3.3. Melt pool depression

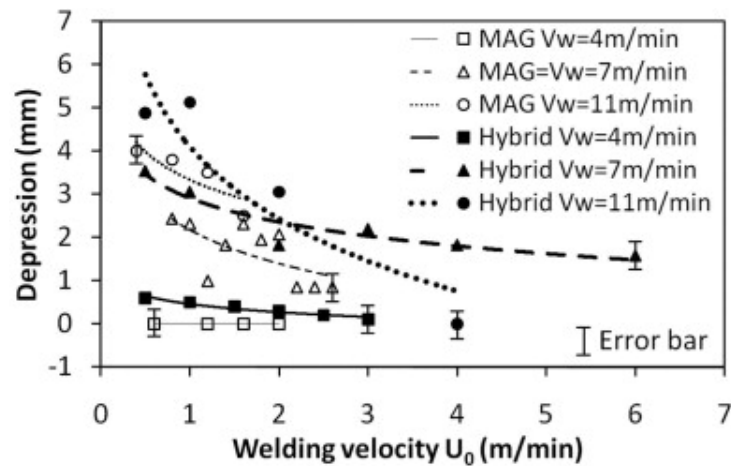
Fig. 8 shows an enlarged view of the depressed zone below the arc during MAG processing. One can clearly see that under the arc centreline the melt pool surface is depressed below the initial work piece surface by more than 3mm. The level of this depression depends on the intensity of the arc current and begins to become significant for currents above 150A. This depression results from the arc pressure due to the induced hydrodynamic gas flow and of course of the impingement of melt droplets. It is clear that the thermal field induced inside the work piece by the arc from this depressed surface has to be very different from the one obtained if the arc is deposited on a flat surface. The characterization of this depression is, therefore, very important for its comparison with corresponding simulations. The evolution of the depression “ h ” along the welding direction, in the symmetry plane of the weld seam, can be very easily determined from the video images by taking into account the video perspective and the inclination angle of the camera axis with the solid surface of the sample by using the simple relation $h=h'/\cos(45^\circ)$, where h' is the projected value of the depression measured on the recorded image plane (see an example in Fig. 8). The uncertainty in the measurement of the depression h' is ± 0.2 mm. This uncertainty is obtained by making several measurements

of depression on the same video. An error of ± 0.2 mm on the measurement of depression leads to a numerical error on the maximum arc pressure of only ± 50 Pa. Fig. 9 shows the variation of these maximum obtained depressions, as a function of the welding speed U_0 , for different wire feeding rates. One must also add that the depressions determined by this method are in good agreement with those observed directly on the work piece after welding, at the end of the weld seam. One can easily understand that the depression increases for higher wire feeding rates, due to higher arc current, and for lower welding speeds U_0 , due to resulting larger melt pool widths. It is important to note that this rather high value of the induced depression is typically half of the final obtained melt pool penetration; therefore, the effect of surface depression must be taken into account for a correct simulation of the experimental melt pool geometries. We also note that the hybrid depressions are slightly larger than the depressions obtained with MAG alone.



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Fig. 8. Visualization of depression below the arc for the MAG process (incident arc power $UI=8$ kW). The dotted line shows the symmetry line of the weld seam located on the flat surface of the sample and the continuous line shows the profile of the depressed zone. Dimension h' is the projection of the real depression on the observation plane, which is perpendicular to the video camera axis.



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Fig. 9. Maximum surface depression below the solid surface for MAG and Hybrid processes, as a function of welding speed for different wire feeding rates V_w (hybrid parameters: P_{laser} : 4kW, wire diameter: 1.2mm, distance wire/beam: 2mm).

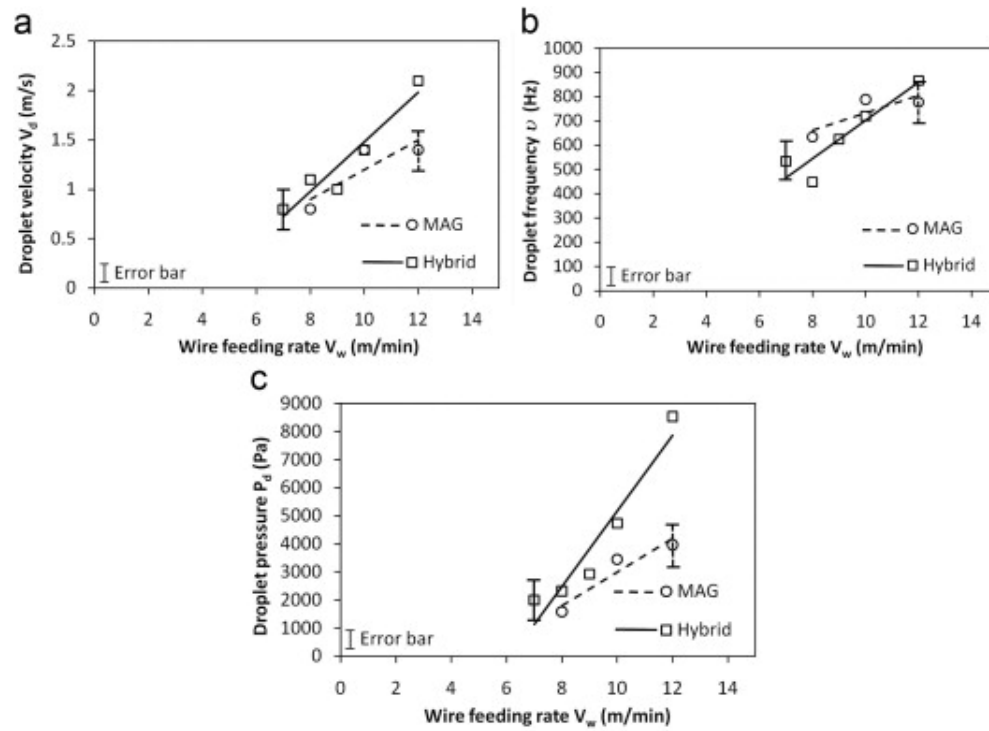
3.4. Droplets pressure measurements

In the spray mode, which is the main metal transfer mode of our experiments, the droplets impinging at rather high velocity the melt pool surface must also have a non-negligible contribution to the melt pool surface depression. If one considers that m_d is the mean mass of a droplet, V_d its velocity, and R_d its mean radius (about 0.4mm), the resulting pressure P_d obtained during an impact of droplets emitted with a frequency ν is given by:

$$P_d = \frac{\nu m_d V_d}{\pi R_d^2} \quad (2)$$

In Eq. (2), we have considered that there is no spreading of the droplet during its impact onto the surface. These different parameters can be easily measured from videos.

Fig. 10(a–c) shows, respectively, the droplet velocity, droplet frequency and the resulting induced pressure (given by Eq. (2)) for the hybrid process, as a function of the welding speed and for different wire feeding rates. As there is some acceleration of the droplet due to the drag force of the plasma flow, the droplet velocity reported in Fig. 10a is an average velocity along its trajectory. As the wire feeding rate increases, it is expected that the droplet velocity and the frequency of detachment increase (as shown by the experimental tendency). There is no clear effect of the welding speed and the frequency of detachment ν , which varies in the range from about 400 to 1000Hz (Fig. 10b). These values of frequencies are in good agreement with previous values observed by Valensi [23]. He studied the influence of oxygen in the shielding gas on the droplet frequency. For a wire with a high content of silicium, as in our case, this frequency increases from 0 to 750Hz when the oxygen level decreases from 70% to 0% and is equal to 700Hz at 8% oxygen as in our study. Measurements of size, velocity and frequency of the droplets are also in agreement with the work of Lin et al. [24]. Indeed, droplet diameters were found around 2–3mm in globular and 1 mm in spray. The droplet frequency increases from 50 to 500Hz when the wire speed (or arc intensity) increases from 7 to 12m/min for a wire diameter of 0.9mm. The droplet velocity in this work increases by 0.5–1.4m/s when the current increases from 160 to 150A for a wire diameter of 1.2mm.



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Fig. 10. For hybrid laser-MAG process, (a) droplet velocity, (b) frequency and (c) induced pressure (given by Eq. (1)), as a function of the wire feeding rates.

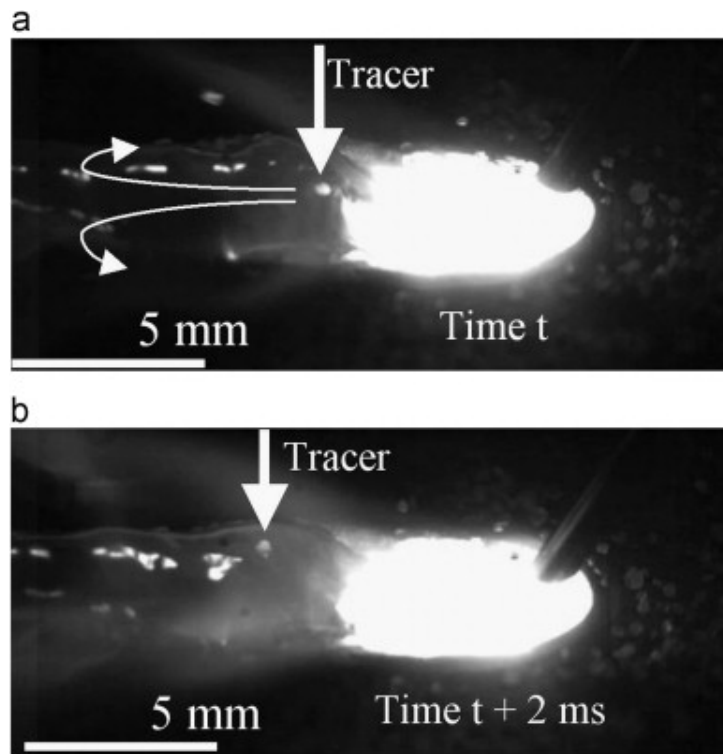
The induced droplet pressure varies from 2 to 8 kPa; as expected, it increases with the wire feeding rate that controls the arc current. These pressures are in agreement with the results of Xu and Wu [25], which predicts zero droplet pressure at low intensity (low wire feeding rate) and pressures up to 5 kPa for the high intensity. Even by taking into account some increase in the impact diameter due to local spreading during the impact, it appears that these values are not negligible compared to the arc pressure that we will determine in Section 4.2. Therefore, the droplet impact must be taken into account as a very local contribution of pressure on the melt pool surface. This impact induces a localised penetration that can be observed, at high wire feeding rates, by a pronounced penetration

on some macrographs or deep deformation of the melt pool surface observed in videos (see [section 4.2.1](#)).

3.5. Melt pool surface speed

From the videos, one can see that these melt pools have complex hydrodynamic flows. So as to estimate the importance of convection and conduction processes inside these flows for energy transport, it would be necessary to determine the fluid flow field. This will dictate the necessity of taking into account, the complete description of the hydrodynamic flow field in the simulations. In order to estimate the magnitude of fluid speed at the melt pool surface, tracer particles were deposited on the sample surface before welding. During melting, these tracers follow the fluid motion. It is then possible to track the motion of particles from the video sequences and deduce fluid velocities by assuming that the particles faithfully follow the fluid dynamics. Different tracers were tested, and it appeared that small silica grains (about 0.1 mm in diameter) had the best behaviour for the MAG and hybrid processes and could be easily followed on the videos.

[Fig. 11](#)(a,b) show 2 images extracted from a video sequence where the displacement of the tracer is observable during the corresponding elapsed time between these 2 images.

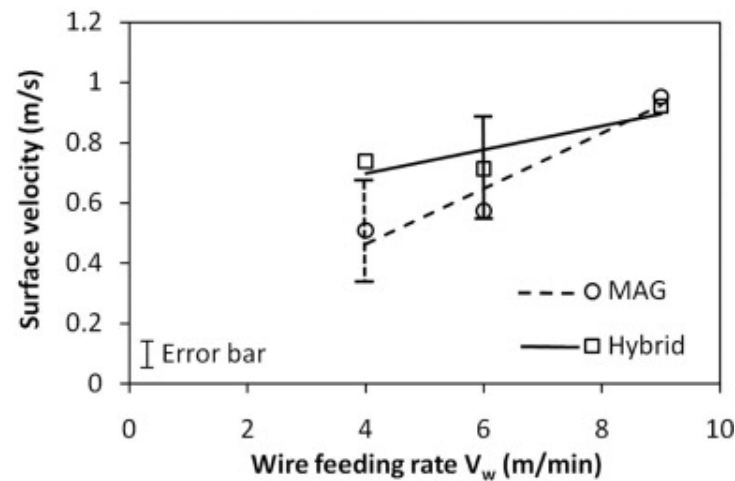


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Fig. 11. (a,b): two images extracted from a video sequence showing the displacement of a tracer (indicated by an arrow) between a time interval of 2ms (MAG process: welding speed U_0 : 2 m/min; wire speed V_w : 9m/min). General trend of the melt flow surface is also indicated by curved arrows in Fig. 11a.

Fig. 12 shows typical tracer velocities observed at the centre of the melt pool behind the arc, along the transition from the arc depression to the weld reinforcement. It is at this position that the surface speed appears to be maximum; then, behind that zone, the surface flow is redirected towards the edges of the melt pool with a vortex-like structure (see arrows in Fig. 11a). This effect is rather similar for the hybrid and MAG processes. The measurements show that the fluid speed increases with the wire feeding rate. This speed is quite constant with the welding speed and is slightly greater in hybrid than in MAG at

low currents. The presence of the keyhole modifies the local fluid motion since the fluid has to flow around the keyhole. In addition, the vapour ejected from the keyhole modifies the fluid motion dynamics [15]. The magnitude of maximum surface velocities results in being rather high, in the range 0.5–1 m/s, and so correspondingly the Peclet number (Pe) that can be estimated as $Pe \sim 200\text{--}300$. This would indicate that the convective transfer of energy is not negligible compared to the conductive process. However, these velocities are measured at the surface of the melt pool, and may largely overestimate the real internal flow speed.



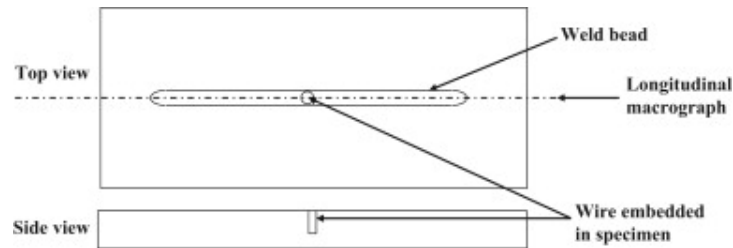
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Fig. 12. Tracer speed as a function of welding speed for different wire feeding rates for hybrid and MAG processes at a typical welding speed of 1 m/min.

3.6. Analysis of melt pool geometry along the welding direction

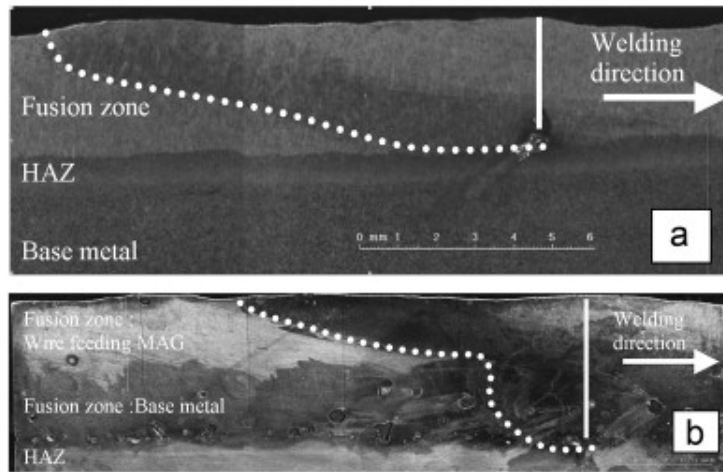
The 3D shape of the melt pool inside the work piece, during the process is also an interesting parameter for modelling comparisons. Analysis of longitudinal sections is more sensitive, in order to determine the contour of the melt pool, which is not normally visible due to the continuity of the melt pool in the solidified zone. For an estimation of the longitudinal shape of the melt zone, we use a small amount of contrast material,

which is different from the S355 base material and filler wire, which is inserted perpendicularly to the work piece. This contrast material is introduced before welding, as a wire inside a hole drilled perpendicularly to the work piece, on the symmetry plane of the weld seam (see Fig. 13). If its diameter is small enough, it does not disturb the melt pool hydrodynamics; this point is verified through the width and length weld bead measurements. When the melt pool reached the contrast material, it is assumed that the contrast material is redistributed very quickly throughout the available volume of liquid before its solidification. Then chemical etching can be adapted to reveal the contour, as can be seen in longitudinal sections (Fig. 14). Different contrast materials and wire diameters have been tested and it appears that a copper alloy (Cu-10% Al) wire with a diameter of 1 mm gives the best results.



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Fig. 13. Scheme using the tracer implant inside the work piece.



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Fig. 14. Longitudinal macrographs obtained with copper/aluminium alloy contrast material. The dotted line shows the melt pool boundary. Vertical white bar shows the initial position of the contrast wire. (a) MAG alone ($U_0=0,8\text{m/min}$, $V_w=5\text{m/min}$) and (b) hybrid laser/MAG ($P_{laser}=4\text{kW}$, $U_0=1\text{m/min}$, $V_w=5\text{m/min}$, wire/laser beam distance: 2mm).

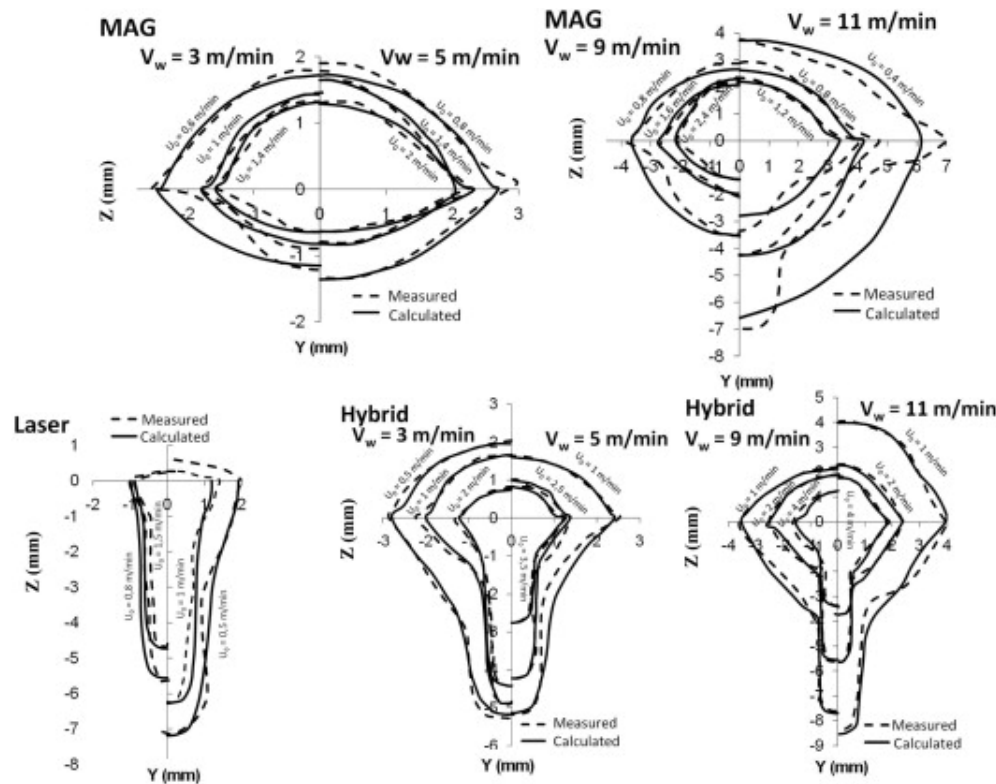
However, this technique has some limitations; it can be used only for configurations with rather low MAG filler wire rates; for high power MAG, thus correspondingly high filler wire inputs, the contrast with the copper alloy wire is not sufficient. Nevertheless, when this technique can be applied, interesting melt pool longitudinal internal profiles can be obtained (see Fig. 14). They will be discussed and compared with corresponding simulations in Section 4.2. As expected, for the MAG process, the internal melt pool has a shape of a rather flat bowl (Fig. 14a). But in case of the hybrid process (Fig. 14b), this shape is modified by the localised melt flow occurring around the deeply penetrating keyhole.

4. Numerical simulation

4.1. 3D thermal numerical model

In order to analyse these experimental observations by using simulations, one must take into account the addition of the filler wire and the observed depression resulting from the arc pressure and the momentum of the droplets flow previously described experimentally. It necessitates describing more precisely the interaction zone and it is also intended to estimate the weld reinforcement and root drop out for joints with a gap. Despite the rather high value of the characteristic Peclet number inside the melt pool we have previously determined in [Section 3.4.](#), the melt pool hydrodynamics was not taken into account in these simulations. This is, of course, a great simplification for these simulations and we will see in [Section 4.2.](#) that the use of a modified thermal conductivity inside the melt pool allows us to correctly reproduce its size.

For the description of the laser keyhole welding alone and also for describing the laser interaction for the hybrid process, we used a previous analysis of Lankalapalli et al. [\[26\]](#) where they consider that the keyhole is a vertical cylinder with a uniform temperature along its wall that is slightly above the evaporation temperature of the material involved. Because of the rather low welding speeds used here for these hybrid welding conditions, we know that the keyhole is almost vertical [\[15\]](#). However, concerning the hypothesis of the keyhole wall temperature close to the evaporation temperature of the material, we know, since the study of Semak and Matsunawa [\[27\]](#), that this hypothesis is not necessarily verified during laser welding. But nevertheless we have used it, because of its simplified consequences in terms of modelling and its ability to easily reproduce the geometry of the longitudinal and transverse weld seam ([Fig. 15](#)). Based on experimental observations, the keyhole diameter corresponds to the focal spot diameter (0.6mm). The depth of the keyhole can then be defined when the integrated heat flux through the entire keyhole wall surface corresponds to the total absorbed laser power. An analytical description can also be used for that determination [\[26\]](#).



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Fig. 15. Comparison between measured and calculated cross section weld seam for MAG, laser and hybrid processes.

For the arc and hybrid description, the high speed videos have shown the shape of the interaction zone: a depression of the melt pool under the arc followed by weld reinforcement due to the filler wire. In that case, the weld pool surface is not flat and its final shape must satisfy an equilibrium condition between surface tension, hydrostatic pressure and arc pressure.

Ushio and Wu [20] have already proposed a solution for solving this problem taking into account these constraints, on the basis of an approach that minimises the total energy of the melt pool, by adding the constraints that the melt pool has a constant volume and the

weld reinforcement geometry is equal to the filler wire mass flow. This total energy of the melt pool taken into account concerns the variation of surface energy with a corresponding change of its area, the total potential energy inside the gravitational field, and the work performed during that change of area by the arc pressure and droplet flow momentum. The resulting equation describing the height of the surface profile $\Phi(x,y)$ is given by Eq. (3) where x and y correspond, respectively, to longitudinal (or welding) and transverse directions, in the laser beam frame:

$$-\sigma \left[\frac{(1+\phi_y^2)\phi_{xx} - 2\phi_x\phi_y\phi_{xy} + (1+\phi_x^2)\phi_{yy}}{(1+\phi_x^2+\phi_y^2)^{3/2}} \right] = -\rho g\phi + P_{surf} + C \quad (3)$$

$$P_{surf}(x, y) = P_{max} \exp \left(-\frac{(x^2+y^2)}{2r_1^2} \right) \quad (4)$$

where σ is the surface tension, ρ the mass density, g the gravity, $P_{surf}(x,y)$ the applied pressure distribution at the melt pool surface, where P_{max} is the maximum pressure with a characteristic radius r_1 . For the sake of simplicity, we have combined in Eq. (4) the effect of arc and droplets. The index x and y are the respective derivative of profile $\Phi(x,y)$ with x and y variables. C is a Lagrange Multiplier adjusted by the mass conservation constraint of the feeding wire that defines the transverse profile shape $\Phi(x,y)$ of the solidified weld seam that must fulfil the following equation:

$$\pi r_w^2 V_w = U_0 \int (\phi - z_0) dy \quad (5)$$

where r_w and V_w are the wire radius and feeding speed, U_0 the welding speed and z_0 the height of the initial work piece flat surface.

$$\Phi_{surf}(x, y) = \frac{\eta UI}{2\pi r_0^2} \exp \left(-\frac{(x^2+y^2)}{2r_0^2} \right) \quad (6)$$

Eq. (6) defines the heat flux $\Phi_{surf}(x,y)$ deposited by the MAG on the melt pool surface. The arc efficiency of the process η and the characteristic radius r_0 are adjusted in order to reproduce the experimental results concerning the melt pool geometry (width, depth). A Gaussian distribution was chosen to describe the arc power. This kind of distribution is

widely used in the literature [20], [28], [29], [30], [31]. However, Xu et al. [32] recently highlighted that the heat distribution is far from being symmetrical. He directly calculated the physical processes inside the plasma arc and showed that the arc pressure and energy distribution are not Gaussian. Indeed, the distribution is altered by the effect of droplet detachment. In our study, as we do not take into account the description of the arc plasma and the interaction of droplets with the melt pool, a simplified Gaussian distribution for the power deposition flux was chosen. Arc pressure distribution defined by Eq. (4) and the maximum pressure are adjusted in order to reproduce the melt pool depression determined experimentally from high speed videos. More detailed description concerning the heat equation and the algorithm used for solving the complete set of equations by using the Comsol software [33] will be given in a companion paper [34].

4.2. Results and discussion

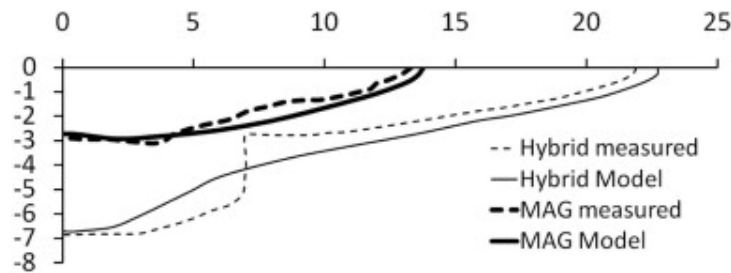
Using the previous procedure, we have simulated 28 experimental tests: 4 tests concerned laser alone, 12 the MAG alone and 12 the hybrid process. These experiments concerned only fusion lines; however, we found that these configurations are similar to assemblies with a gap width of less than 200 μm . For these experimental tests, the welding speed and the filler wire rate were the main parameters that have been varied. Only four unknown parameters of Eqs. (4), (6) (P_{max} , η , r_0 and modified thermal conductivity) have been determined by comparison with experimental data given by weld seam cross section and melt pool (geometry and depression) analysis. In the following sections, we will discuss the methodology used for deriving these parameters and the physical mechanisms that can be inferred from these analyses.

4.2.1. Arc efficiency and intensity distribution; heat conductivity modification

Fig. 15 shows the comparisons between experimental and numerical results concerning the weld seam cross section, for the 3 processes. Generally, a rather good agreement is obtained between numerical and experimental cross sections. However, these

comparisons show some limits to our approach: Indeed, in the case of MAG welding, at high filler wire speed (11 m/min) and low welding speed (0.4 m/min), the observed complex geometry, similar to that observed in hybrid welding and probably resulting from very localised intense droplet impingement, cannot be correctly reproduced using this purely conductive model. However, the model reproduces a wide operating range corresponding to the configurations currently used in industry.

It is important to note that another parameter, the liquid thermal conductivity, had to be modified in order to correctly reproduce the melt pool length (see [Fig. 16](#)). In fact, this liquid heat conductivity had to be increased by a factor K , depending on the experimental conditions that point out some physical processes occurring inside the melt pool. For example, in order to reproduce the melt pool length in the MAG process alone (otherwise it was twice as long), the factor K had to be set to about 3. This isotropic increase of liquid thermal conductivity has the effect of homogenizing the temperature field, which is what is expected by a hydrodynamic flow inside the melt pool, which is not taken into account here. Moreover, this increased liquid thermal conductivity also has the effect of reducing the maximum surface temperature to a level near the evaporation temperature of the material. Otherwise, for these input surface intensities, and welding speeds, the obtained temperatures would have been much higher, which is not physically possible. For a similar approach in the case of the laser process alone, an anisotropic conductivity in the welding direction has to be introduced. This conductivity in the welding x -direction had to be multiplied by a typical factor K of 5–6 compared to the transverse one in order to reproduce the experimental weld pool length. Without this increase, the simulated weld pool was much shorter compared to the experimental one. The high value of this anisotropic K used here probably reflects the strong flow field obtained in laser welding compared to MAG conditions.



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Fig. 16. Comparison of experimental and calculated longitudinal weld pool shape for: MAG alone: welding speed: 1 m/min, wire speed: 9 m/min, current: 200 A, voltage: 17.6 V and hybrid Laser-MAG: welding speed: 1 m/min, Laser power: 4 kW, wire speed: 5 m/min, current: 200 A, voltage: 17.6 V, distance wire/beam 2 mm.

Finally for hybrid welding, the K factor used had to be varied from 3 (this increase being in that case isotropic) for high filler wire rates (corresponding to large melt pools) to an anisotropic K of about 5–6, at low filler wire rates, when the arc power was less important. This K variation is consistent with the expected evolution of the hydrodynamic melt pool flow as a function of the operating parameters. An example of a longitudinal melt pool shape for the hybrid welding case is shown in Fig. 16. The melt flow localized around the penetrating keyhole below the melt pool generated by the arc is not correctly reproduced because only the thermal field is simulated here, without taking into account the hydrodynamics of the melt pool. Table 1 shows a summary of thermal conductivity modifications with the various welding configurations.

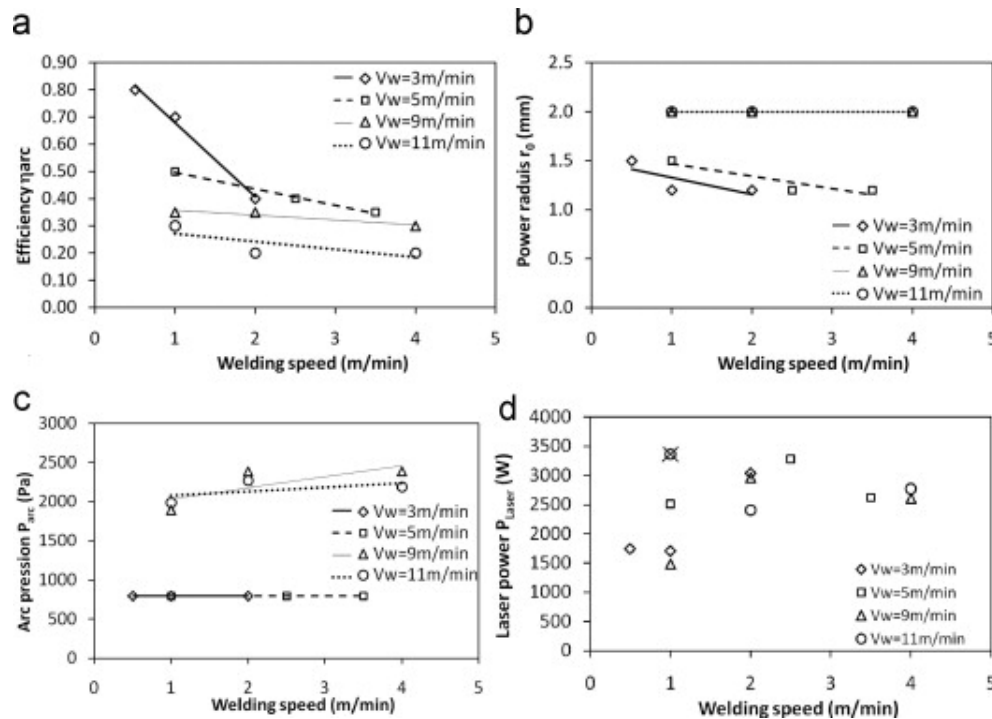
Table 1. Summary of liquid thermal conductivity adjustments used for the various configurations.

Laser	Anisotropic, along welding direction: $\times 6$	
	$2 \text{ m/min} < V_w < 7 \text{ m/min}$	$7 \text{ m/min} < V_w < 12 \text{ m/min}$

Laser	Anisotropic, along welding direction: ×6	
MAG	No modification	Isotropic: ×3
Hybrid laser/MAG	Anisotropic, along welding direction: ×6	Isotropic: ×3

It is also important to add that during this identification procedure of these unknown parameters, we have observed that the determination of the couple of unknown parameters, such as the arc efficiency η and the radius r_0 of power deposition cannot be unique if one analyses only a weld seam cross section. It is only when the melt pool length is also considered that a unique determination of the couple (η and r_0) is possible.

The arc efficiencies η determined by using these simulations have been reported in [Fig. 17a](#) as a function of welding speed for different filler wire speeds. We recall that the efficiency we defined here corresponds to the fraction of arc power UI that is used for reproducing the observed geometry of melt pool and final weld seam cross section. For these simulations, the only loss mechanisms used were the standard losses from convective exchange with ambient air and from grey body emissivity. One can see that this arc efficiency does not quite depend on the welding speed but it increases from 30% in spray regime (high arc power, about 8–10kW) to 70% in short-circuit regime (low arc power, about 3–4kW). The high efficiency for short-circuit regimes can be easily understood. For high arc current regime (spray mode), radiation losses from the arc is a possible reason for such reduced efficiency. This tendency was also observed in TIG welding by Niles and Jackson [\[35\]](#). Their measurements have indicated that the arc efficiency decreased from 60% to 30% when the intensity increased from 125 to 225A. We may also add that evaporation losses were not taken into account here. At high arc current it is experimentally observed a high level of evaporation from the melt pool surface due to the high resulting surface temperature. Moreover, considerable spattering was observed for these high arc current regimes.



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Fig. 17. Parameters used in the simulation to reproduce the weld pool characteristics determined from macrographs and melt pool length measured using videos: (a) arc efficiency, (b) arc power radius, (c) arc pressure and (d) laser power, obtained with hybrid laser-MAG for different wire speeds.

The precise estimation of these evaporation losses is difficult because the level of re-condensation of ejected metallic vapours occurring at atmospheric pressure is not known. If one uses the evaporation rate given by the Langmuir law [36], [37], we observe that losses are largely overestimated and cannot be included at this level. DebRoy and David [37] have pointed out that this law can be used for evaporation rates at a low pressure environment when no significant re-condensation occurs. His experiments indicate that evaporation rates in usual welding conditions would be 5–10 times lower than those predicted by the Langmuir law. Thus, for these reasons, we did not take into account these

evaporation mechanisms in the simulations, because of these very large uncertainties on these mechanisms. One must also add that even if evaporation losses are efficient for reducing surface temperature to levels near evaporation temperature, they were not able to reproduce experimental melt pool lengths, as the increased liquid heat conductivity did, with a similar effect on surface temperature. The radius r_0 used in Eq. (6) for characterizing the profile of deposited arc power has been reported in Fig. 17b. As expected, this radius increases with the arc current, and as for arc efficiency, this parameter is not sensitive to the welding speed.

4.2.2. Laser efficiency

Fig. 17d shows the laser power that must be used in the simulations in order to reproduce the experimentally obtained keyhole depths, as the function of the operating parameters. There is a singular point at $V_w=11$ m/min and $U_0=1$ m/min, which is not consistent with the other results that show a general increasing tendency of the useful laser power with the welding speed. In fact, for MAG welding alone, at high filler wire speeds and low welding speeds, we have already seen in Fig. 15 a similar deviation, with a rather deep penetration that had already been observed. Thus for corresponding hybrid welding conditions, a crude analysis considering that this observed deep penetration results only from the laser keyhole is not correct and thus this point has to be discarded.

As the incident laser power was set to 4kW, one can see that maximum necessary useful fraction reaches about 75% and in spite of some dispersion of the data, it is clear that the laser efficiency decreases when the welding speed decreases. This behaviour has already been analysed for laser welding alone [22]. In fact, this result can be explained if one considers that low welding velocities induce deep keyholes, with large depth/diameter ratios, the keyhole geometry then becomes unstable due to the Rayleigh–Taylor instability [22]. As a consequence of these internal fluctuating closures, the keyhole is non-stationary, thus a stationary thermal model is not appropriate for this description: as the resulting penetration depth is limited by this instability, the corresponding laser power necessary for reproducing this penetration depth is of course reduced. As this

behaviour becomes dominant at low welding speeds, the effective laser efficiency can be much lower than the usual 80% to 90% observed laser efficiency resulting from the geometrical trapping of laser beam inside the keyhole. Thus laser efficiency is a rather sensitive function of the welding speed, particularly in case of hybrid welding where the keyhole can be easily perturbed by the rather large surrounding melt pool generated by the arc.

4.2.3. Arc pressure determination

The induced pressure applied at the melt pool surface is given by Eq. (4) by defining the two parameters P_{max} and r_1 . In fact, unlike the determination of the arc power parameters, it is not possible to show the uniqueness of the couple (P_{max}, r_1) using only the maximum depression measurement, because several couples (P_{max}, r_1) can give the same maximum depression, but with significantly different profiles. Therefore, again to simplify the analysis by minimizing the number of parameters to determine, we have chosen to use the radius of pressure distribution r_1 that is equal to the radius distribution r_0 . Thus, the maximum pressure has been adjusted to obtain the maximum observed depression.

The arc pressure evolution for hybrid conditions is reported in Fig. 17c. The uncertainty on the arc pressure of about 100Pa, results from the accuracy on depression measurements, which is about 0.3mm. The maximum pressure P_{max} (Eq. (4)) clearly increases from the short circuit regime with pressures of about 800Pa to the spray regime with pressures of about 2300Pa. We have also observed that pressures for hybrid conditions are about 30% higher than for the MAG process. These results can be understood if one considers that the arc pressure is directly related to the arc current. Finally, one can notice that these pressures determined by this indirect method are in agreement with those determined by Lin and Eager [38].

5. Summary of the experimental results and corresponding numerical parameters

The large amount of the experimental data presented in [Section 3](#), and the corresponding analysis of determined parameters for reproducing them by using our numerical approach, discussed in [Section 4](#), has allowed us to obtain rather precise analytical expressions of these different parameters as a function of the 2 main operating parameters that were used in these experiments, which are the welding speed U_0 and the arc power P_{arc} (P_{arc} is defined by giving the filler wire speed V_w used, and the corresponding arc synergy defined in [Fig. 3](#)). These analytical expressions have been obtained by determining a best fit of a polynomial expression of the 3rd degree with the variables U_0 and P_{arc} , with the corresponding results. We used the following functions:

$$Y(U_0, P_{arc}) = a_1 U_0^3 + a_2 U_0^2 P_{arc} + a_3 U_0^2 + a_4 U_0 P_{arc}^2 + a_5 U_0 P_{arc} + a_6 U_0 + a_7 P_{arc}^3 + a_8 P_{arc}^2 + a_9 P_{arc} + a_{10} \quad (7)$$

where the constants a_1 to a_{10} are given in [Table 2](#), [Table 3](#) for the different data. The mean accuracy of this polynomial fit with the corresponding results, given in the last column in these two tables, is rather correct. These determinations are valid on our range of the operating parameters used: from 0.4 to 4m/min for the welding speed U_0 , and from 2 to 11 kW for the arc power P_{arc} . These analytical functions are very useful because for any set of operating parameters U_0 and P_{arc} given in the previous validity range, one can obtain with [Table 2](#) the corresponding experimental geometrical parameters of the resulting melt pool, and with [Table 3](#), the numerical parameters necessary to use in order to reproduce this melt pool with our simulations.

Table 2. For hybrid welding conditions, polynomial expressions for several experimental geometrical melt pool parameters (U_0 is expressed in m/min and P_{arc} in kW).

Parameter	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	a_9	a_{10}	(%)
Melt pool width (mm)	-0.16	0.07	1.5	0.0017	-0.51	-4.23	-0.0024	-0.01	1.41	5.37	11
Penetration depth (mm)	0	0	0.14	0	-0.078	-1.45	0	-0.0079	0.52	5.65	7
Weld seam height (mm)	-0.2	-0.0091	0.31	-0.0032	0.064	-1.47	0.0044	-0.06	0.36	2.07	12
Melt pool depression (mm)	0	0	0.12	0	-0.10	-0.47	0	0.0039	0.63	-0.72	24
Melt pool length (mm)	0	0	0.19	0	-0.13	-1.11	0	-0.0661	3.70	11.9	8

Table 3. For hybrid conditions, polynomial expressions for the parameters used in the numerical simulations, (U_0 is expressed in m/min and P_{arc} in kW).

Parameter:	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	a_9	a_{10}	(%)
Arc pressure (Pa)	35.3	-22.4	-169.2	-5.5	172.3	125.3	-11	209.6	-1128	2330.2	2
Arc efficiency (%)	-2.78	-0.16	22.289	-0.41	7.27	-79.5	-0.1	2.62	-26.054	156.8	7
Arc radius (mm)	-0.019	-0.004	0.1708	-0.0011	0.05	-0.65	-0.0033	0.5	-0.14	1.79	2.5
Laser efficiency (%)	-7.778	3.6	21.47	-1.45	3.28	-14.4	0.51	-7.0	34.75	-7.1	10

6. Conclusion

A set of experimental observations has been realized for a wide range of the main operating parameters concerning the hybrid laser-MAG process. The main geometrical characteristics of the melt pools have been determined and we have seen the importance of melt pool depressions as a result of applied pressure at the surface of the melt pool due to arc interaction. By using numerical simulations, we have reproduced the final cross section of the weld seams and the 3D shapes of these melt pools. For the last point, we have shown the interest of increasing the liquid thermal heat conductivity in order to simulate heat hydrodynamic flow inside the melt pool. This procedure has allowed us to determine the corresponding parameters related to arc and laser energy deposition and also the resulting applied pressures at the melt pool surface, as a function of the used operating parameters. This approach can be applied to an industrial configuration (with a groove, or for full penetration). This paper presents encouraging results concerning the modelling of the hybrid laser-MIG/MAG process obtained by using rather simple approaches and hypothesis that allowed us to reproduce realistic melt pool geometries.

7. Future work

With these simulations, we should now be able to estimate any temperature profile inside the work piece. Future experiments will concern direct measurements of temperature profiles (by using embedded thermocouples or thermal camera) in order to validate the temperature profiles predicted by these simulations and to estimate the corresponding metallurgical transformations. New steps of this program should also consist of describing hydrodynamic flow of the melt pool and thermocapillary effects, for studying their consequences on geometry and energy transport inside the melt pool order to reproduce the melt pool in MAG welding for extreme configurations (high wire speed and low welding speed). This much more complex task should be possible by using the same Comsol software.

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